

Hong Yuan Cao

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EDUCATION

Boston University

Boston, MA

Double Major: Bachelor of Arts in Computer Science and Economics, GPA: 3.74/4.0

Graduation: May 2026

Relevant Courses: AI, ML, Programming Languages, Distributed Systems, Databases, Software Engineering, Analysis of Algorithms, Computer Systems, Probability in Computing, Combinatoric Structures, Discrete Mathematics, Data Structures

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, Go, C, C#, SQL, OCaml, HTML, CSS, Shell, LaTeX, Assembly
Frameworks: Django, React, React Native, Vue.js, Electron, Flask, Node.js, Spring, PyTorch, Pandas, OpenCV, gRPC
Tools: Git, Docker, Unix, Linux, Raspberry Pi, MySQL, PostgreSQL, Redis, AWS, ChromaDB, Terraform, GDB

EXPERIENCES

Software Engineer Intern

Shanghai, China

KeelinMicro Electronics (Automotive Startup)

June 2025 – August 2025

- Co-invented a **patent-pending** CV architecture achieving **96.3%** object tracking continuity and **31%** power reduction
- Engineered an **OpenCV** pipeline with Python to generate 3D point clouds of objects from live video at over **60 FPS**
- Architected a real-time synchronization system for camera and radar inputs, achieving **sub-millisecond accuracy**
- Built custom **Linux kernel driver** in C and **device tree overlay** to enable GMSL camera streaming on Raspberry Pi 5

Software Engineer Intern

Shanghai, China

Tospur Real Estate Consulting Co., Ltd

June 2024 – August 2024

- Built **Django ETL pipeline** on Alibaba Cloud consolidating several databases, reducing data aggregation time by **45%**
- Migrated **200+** consultants from 4 legacy systems to unified platform by implementing secure credential migration APIs
- Optimized database performance via **query caching** and **connection pooling**, reducing query latency by **23%**

Machine Learning Intern

Shanghai, China

Anban Tech (Fuzz Testing Startup)

May 2024 - June 2024

- Built preprocessing pipeline to clean code-testing datasets, improving data quality for ML model training using **Pandas**
- Curated **2.2GB JUnit 5** dataset from open-source repositories using **GitHub API** for fuzzing LLM training
- Developed **Python** validation scripts achieving **95%** accuracy of dataset format validation for training pipeline

Software Engineer Intern

Shanghai, China

Tonic – Sprout Technologies, Inc. (Entertainment Tech Startup)

June 2023 – October 2023

- Implemented responsive UI components and animations using **React.js**, increasing landing page engagement by **24%**
- Built a social media crawler using **Python (BeautifulSoup4)** that identified **200+** qualified influencers to target
- Filtered **3,000+** media contacts using **Python (Pandas)** and Excel, improving outreach response rates by **25%**

PROJECTS

Athens - AI Learning Platform | AWS, FastAPI, React, TypeScript, OpenAI, PostgreSQL

- Engineered scalable **AWS architecture** with **ECS Fargate**, **RDS PostgreSQL (pgvector)**, **S3**, and **CloudFront**
- Developed end-to-end **RAG** system with **pgvector** embeddings, processing **1000+** documents with semantic retrieval
- Built **FastAPI/React** full-stack with context injection and session management, improving response quality by **40%**

Serverless Image Processing Pipeline | AWS S3, Lambda, CloudFront, DynamoDB, Terraform, React

- Deployed serverless infrastructure using **Terraform**, **S3**, **Lambda**, **CloudFront**, reducing delivery latency by **38%**
- Implemented **Python-based Lambda** functions triggered by **S3** events to automatically resize uploaded images

Multi-Currency Budget Tracking Mobile App | Django, React Native, PostgreSQL, Android Studio

- Developed a expense tracking app with **React Native**, featuring currency conversion via third-party API integration
- Built **Django REST APIs** using **composite indexing** and **ORM optimization**, reducing retrieval by **65%**

Movie Recommendation Platform | Flask, Vue.js, Python, Javascript, PostgreSQL

- Integrated movie database API with **Flask** to create an interactive dashboard using **Vue.js** components
- Built content-based recommendation algorithm using **CountVectorizer** and **cosine similarity** to analyze metadata

Neural Language Model Implementation | Python, PyTorch

- Implemented transformer architecture in **PyTorch** with token and positional embeddings for sequential prediction